

August 29, 2024

Rating:

OUTPERFORM

Price:

\$125.61

12-Month Price Target:

\$138.00

Analysts

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Company Information - Hardware

Market Cap (M) \$3,090,006

Enterprise Value (M) \$2,746,080

Shares Outst (M) 24,848.0

52-Week Range \$39.23 - \$140.76

REV (M)	in \$		
FYE Jan	2024A	2025E	2026E
Q1 Apr	7,192.0A	26,044.0A	37,933.7E
Q2 Jul	13,507.0A	30,040.0A	41,980.2E
Previous		29,991.9E	41,863.8E
Q3 Oct	18,120.0A	32,503.9E	44,993.7E
Previous		32,369.0E	44,600.5E
Q4 Jan	22,103.0A	34,629.8E	46,574.5E
Previous			46,424.1E
Year*	60,922.0A	123,217.7E	171,482.1E
Previous		123,034.7E	170,822.0E

Gross Margin	in %		
FYE Jan	2024A	2025E	2026E
Q1 Apr	66.8%A	78.9%A	75.0%E
Previous			76.0%E
Q2 Jul	71.2%A	75.7%A	75.0%E
Previous		75.5%E	76.0%E
Q3 Oct	75.0%A	75.0%E	75.0%E
Previous		75.7%E	76.0%E
Q4 Jan	76.7%A	74.5%E	75.0%E
Previous		75.9%E	76.0%E
Year*	73.8%A	75.9%E	75.0%E
Previous		76.4%E	76.0%E

EPS	in \$		
FYE Jan	2024A	2025E	2026E
Q1 Apr	0.11A	0.61A	0.84E
Previous			0.85E
Q2 Jul	0.27A	0.68A	0.93E
Previous		0.67E	
Q3 Oct	0.40A	0.73E	0.99E
Previous		0.72E	
Q4 Jan	0.52A	0.76E	1.02E
Previous		0.77E	1.03E
Year*	1.30A	2.78E	3.78E
Previous			3.79E
P/E	96.9x	45.1x	33.2x

Pricing data provided by Thomson Reuters.

*Numbers may not add up due to rounding.

Nvidia (NVDA)

Back in Black(well) - Strong Sales + Limited Delay Concerns

NVDA's call played out much as we previewed:

- The company reported \$30B in sales, with revenues up ~\$4B Q/Q driven by data center sales (in line with our revised estimate) while management guided for \$32.5B in revenues in FQ3, just above our outlook.
- Regarding Blackwell delays, management described a change to the Blackwell mask to improve yields. NVDA now expects shipments to ramp in Q4 with NVDA planning on revenueing several billion dollars of Blackwell this year.
- GMs are still expected to be in the mid-70s for FY2025 (consistent with prior guidance), though arguably management was more vocal in pointing out this forecast implies a step down in margins over the last two quarters tied to the ramp of new products (H200 with its higher memory content and Blackwell).
- Finally, NVDA management indicated that Hopper revenues are expected to grow in 2H'25 implying no airpocket in revenue prior to the Blackwell ramp.

While the stock dropped meaningfully after hours, fundamentally we didn't see anything in numbers or in NVDA's commentary that was concerning or suggested any slowing in NVDA's momentum. In particular, we'd note that management's Blackwell commentary (while less specific than some of the reports that have circulated) generally mirrored the expected narrative. Specifically:

- The delay is expected to be short lived (< one quarter). We'd also note our conversations always have suggested Blackwell's ramp was likely to begin in CQ4'24, with system builders more optimistic about shipments in CQ1'25+.
- Solid Hopper sales should allow NVDA to maintain its sales growth trajectory until Blackwell ramps.

Net, we see no reason to shift our positive outlook on NVDA. As such we are reiterating our OUTPERFORM rating on the stock. And with our forward numbers remaining nearly static (our FY'26 sales lift modestly, but EPS dips by one penny due to our slightly lower GM expectations), we are leaving our price target unchanged.

Valuation - Our PT remains \$138. We continue to calculate our PT by applying a PE multiple of ~36X to our FY'26 EPS estimate of \$3.78 plus net cash of \$1.01/share.

Derivatives

MU - Green Flag - Continued strong demand for GPUs suggests continued accelerated growth in HBM requirements (particularly with content growing as H200 ramps). We see this result as necessarily being positive for all of the DRAM vendors (albeit with Hynix still the largest immediate direct beneficiary of increasing HBM requirements) and even to some extent NAND vendors as memory resources are increasingly allocated towards ramping HBM (rather than future NAND capacity).

SMCI - Green Flag - NVDA's static customer mix (45% CSP and 55% other) and commentary around sovereign demand strength suggests server OEMs should benefit from solid growth outside of the large CSPs. We see this result as positive for SMCI, Dell, Gigabyte etc., though we would also note that SMCI's near-term stock momentum is more likely to be dictated by the company's delayed 10K.

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NVDA FQ2'25 Review

Data Center– Data Center posted another new record in revenues of \$26.3 billion, increasing 16% Q/Q and 154% Y/Y (from \$22.6 billion in FQ1).

- Compute revenues were \$22.6 billion, increasing 17% Q/Q, and 162% Y/Y (vs. \$19.4 billion, increasing 29% Q/Q, and 478% Y/Y in FQ1), driven by demand for NVIDIA's Hopper GPU computing platform for training and inferencing large language models.
- Data Center growth was once again led by Enterprise and Consumer Internet companies on a Q/Q basis.
- Large CSPs again represented a mid-40s percentage of Data Center revenues, similar to last quarter but a decline from the previous few quarters when NVDA noted CSPs as having contributed more than 50% of Data Center revenues.
- Networking revenues were \$3.7 billion, increasing 16% Q/Q and 114% Y/Y (vs. \$3.2 billion, decreasing 5% Q/Q (due to component constraints), but up 242% Y/Y in FQ1) driven by InfiniBand and Ethernet for AI revenue, including Spectrum-X end-to-end ethernet platform.
- NVDA estimates that in the LTM, 40% of Data Center revenue was tied to AI inference (vs. 40% in FQ1 and FQ4).

Gaming – Gaming results were \$2.9 billion, 9% Q/Q, 16% Y/Y (vs. \$2.6 billion, decreasing 8% Q/Q, but increasing 18% Y/Y in FQ1). The Y/Y increase reflects better higher sales of the company's GeForce RTX 40 Series GPUs and game console SOC's.

Automotive – Revenues for this business were \$346 million, 5% Q/Q, and 37% Y/Y (vs. \$329 million, increasing 17% Q/Q, and 11% Y/Y in FQ1). The Q/Q increase was driven by AI Cockpit solutions and self-driving platforms; the Y/Y increase was driven primarily by self-driving platforms.

Professional Visualization – Revenues were \$454 million, 6% Q/Q, 37% Y/Y (vs. \$427 million, decreasing 8% Q/Q, but increasing 45% Y/Y FQ1). The increase was driven by the continued ramp of RTX GPU workstations based on NVDA's Ada architecture.

OEM & Other – Sales in this segment were \$88 million, 13% Q/Q, 33% Y/Y (vs. \$78 million, decreasing 13% Q/Q, but increasing 1% Y/Y in FQ1).

Gross Margin Results – GMs were 75.7%, decreasing 320 bps Q/Q (vs. GMs of 78.9%, increasing 222 bps Q/Q in FQ1); GM contraction was primarily driven by inventory provisions for low-yielding Blackwell material and a higher mix of new products within Data Center.

Opex Results – Opex 12% Q/Q and represented 9.3% of revenues (vs. increasing 13% Q/Q and representing 10% of revenues in FQ1) with opex growth continuing to be outpaced by NVDA's revenue gains yielding incremental operating leverage. The Q/Q increase reflected higher compensation and benefit expenses driven by growth in headcount.

Use of Cash - In FQ2, NVIDIA returned \$7.4 billion to shareholders, or \$7.2 billion in share repurchases and \$246 million in dividends (vs. returning \$7.8 billion to shareholders in FQ1).

FQ3'25 and Beyond Outlook

Revenue is expected to be \$32.5 billion, plus or minus 2 percent. Management expects Q/Q growth in all market platforms.

Management noted:

- **Sovereign AI** - Sovereign AI revenue to reach LDD billions this year (from de minimis levels the previous year).
- **Automotive** – Management expects automotive to be NVDA's largest enterprise vertical within Data Center this year, driving a multibillion revenue opportunity across on-prem and cloud consumption, and expects it to grow as NextGen AV models require more compute.
- **H200** - The H200 platform began ramping in Q2 and shipping to large CSPs, consumer internet and enterprises.
- **Grace Hopper Superchip** – NVDA's Grace Hopper platform is currently shipping in volume.
- **InfiniBand** – InfiniBand continues to be ahead of NVDA's current supply.
- **Spectrum-X** - Spectrum-X is ramping in volume with multiple customers, and is expected to begin a multibillion-dollar product line within a year.
- **Blackwell** - NVIDIA noted a decision to do a second tape-out of Blackwell to improve yields. We'd note this explanation is somewhat different than reports attributing issues to limited or poor yielding substrate supplies and/or an issue with the bridge connecting the two Blackwell die (though we would postulate perhaps a change in the die could make bridging the chips easier).
- **2H Demand** - The company expects to ship several billion dollars in Blackwell revenue in FQ4, and also mentioned that Hopper demand was strong, with shipments expected to increase in 2H'FY25. Net, NVDA doesn't appear to believe there will be any air pocket in demand due to what now appears to be a modest delay in Blackwell shipments.
- **GMs** – Gross margins are expected to be in the range of 75%, plus or minus 50 basis points. Beyond FQ2, and for the remainder of the year, Management expects GMs in the mid-70 percent range.
- **Operating expenses** – are expected to lift to ~\$3.0 billion. For FY2025, opex is forecast to grow in the mid to upper 40% range, increasing from the previous guide for growth in the low 40 percent range. We believe NVDA will look to ramp opex moving forward as opex growth has significantly trailed revenue growth levels in prior quarters.
- **Other income** – is expected to be ~\$350 million.
- **Tax rate** – is forecast to be 17%, plus or minus 100 basis points.
- **Shareholder return** – The company's board approved an additional \$50 billion in share repurchase authorization.

Estimates, Price Target, and Valuation

We are reiterating our OUTPERFORM rating and \$138 target price. We are applying a 36X multiple to our FY'26 EPS estimate of \$3.78 plus net cash of \$1.4/share. We would note 36X is below the ~40X multiple NVDA enjoyed during its last period of sustained data center expansion, but ahead of where it has traded at over the past 5 years on average.

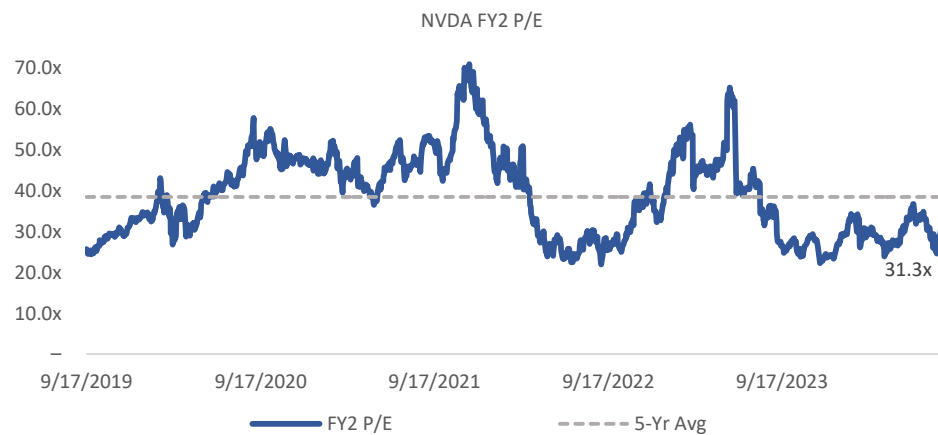
Our new estimates for 3Q25E are: \$0.73, \$32.5B, 75.0%, and 65.8% for EPS, revenues, GM%, and OM%, respectively. This compares with our previous estimates for EPS of \$0.72, revenues of

\$32.4B, GM% of 75.7%, and OM% of 66.0% and previous consensus of \$0.71, revenues of \$31.9B, GM% of 75.4%, and OM% of 65.9%.

Our new estimates for 2025E are: \$2.78, \$123.2B, 75.9%, and 66.5% for EPS, revenues, GM%, and OM%, respectively. This compares with our previous estimates for EPS of \$2.78, revenues of \$123.0B, GM% of 76.4%, and OM% of 66.7% and previous consensus of \$2.76, revenues of \$122.1B, GM% of 76.0%, and OM% of 66.5%.

Our new estimates for 2026E are: \$3.78, \$171.4B, 75.0%, and 65.2% for EPS, revenues, GM%, and OM%, respectively. This compares with our previous estimates for EPS of \$3.79, revenues of \$170.8B, GM% of 76.0%, and OM% of 65.9% and previous consensus of \$3.82, revenues of \$169.3B, GM% of 75.2%, and OM% 66.6%.

Figure 1: NVDA Historical Multiples



Sources: Thomson EIKON and Wedbush Securities

Priced 08/28/2024

Figure 2: NVDA: Updated Wedbush Estimates & Outlook

NVDA (\$M)										
	OUTLOOK	Wedbush	Previous	Consensus	Wedbush	Previous	Consensus	Wedbush	Previous	Consensus
	October-24	3Q25E	3Q25E	3Q25E	2025E	2025E	2025E	2026E	2026E	2026E
EPS	\$0.73	\$0.73	\$0.72	\$0.71	\$2.78	\$2.78	\$2.76	\$3.78	\$3.79	\$3.82
Net Revenue	\$32,500	\$32,504	\$32,369	\$31,913	\$123,218	\$123,035	\$122,061	\$171,448	\$170,822	\$169,305
GM%	75.0%	75.0%	75.7%	75.4%	75.9%	76.4%	76.0%	75.0%	76.0%	75.2%
OM%	65.8%	65.8%	66.0%	65.9%	66.5%	66.7%	66.5%	65.2%	65.9%	66.6%
Segments										
Gaming		3,027	2,999	\$2,938	11,563	11,389	\$11,432	13,056	12,496	\$12,553
Data Center		28,488	28,382	\$28,450	107,897	107,843	\$106,448	153,700	153,538	\$153,080
Data center GPUs		23,979	23,860		92,052	91,403		131,117	130,181	
Mellanox		4,509	4,522		15,845	16,440		22,583	23,358	
Professional visualization		520	520	\$486	1,933	1,953	\$1,855	2,127	2,149	\$2,044
Auto		392	392	\$369	1,488	1,522	\$1,404	2,186	2,282	\$1,903
OEM and other		77	77	\$81	337	328	\$322	379	357	\$349
Total Revenue	\$32,500	\$32,504	\$32,369		\$123,218	\$123,035		\$171,448	\$170,822	
COGS		8,126	7,866		29,752	29,043		42,862	40,997	
Gross profit		24,378	24,503		93,466	93,991		128,586	129,825	
Total Opex	\$3,000	3,003	3,140		11,552	11,909		16,781	17,231	
Operating Income		21,375	21,364		81,915	82,082		111,805	112,593	
Non-op (Expense) Income	\$350	350	302		1,382	1,208		1,400	1,208	
Income tax (benefit) provision		3,693	3,685		14,073	14,167		19,245	19,356	
Tax rate%	17.0%	17%	17%		17%	17%		17%	17%	
Net Income		18,031	17,980		69,223	69,123		93,960	94,445	
Avg. Shares Out - Non-GAAP		24,848	24,890		24,859	24,890		24,848	24,890	
EPS - Non-GAAP		\$0.73	\$0.72		\$2.78	\$2.78		\$3.78	\$3.79	

Sources: NVDA company reports, Thomson EIKON and Wedbush Securities, Inc. estimates

Figure 3: NVDA: Q/Q & Y/Y Comparison

NVDA (\$M)									
	OUTLOOK		Wedbush	Consensus	Var		Q/Q		Y/Y
	July-24	2Q25A	2Q25E	2Q25E	%	1Q25A	%	2Q24A	%
EPS	\$0.62	\$0.68	\$0.67	\$0.64	6%	\$0.61	11%	\$0.27	153%
Net Revenue	\$28,000	\$30,040	\$29,992	\$28,700	5%	\$26,044	15%	\$13,507	122%
GM%	75.5%	75.7%	75.5%	75.8%	-11bps	78.9%	-328bps	71.2%	448bps
OM%		66.4%	65.8%	65.8%	53bps	69.3%	-297bps	57.6%	880bps
Segments									
Gaming		2,880	2,735	\$2,763	4%	2,647	9%	2,486	16%
Data Center		26,272	26,325	\$25,148	4%	22,563	16%	10,323	154%
Data center GPUs		22,604	22,075			19,392	17%	8,623	162%
Mellanox		3,668	4,250			3,171	16%	1,700	116%
Professional visualization		454	474	\$463	-2%	427	6%	379	20%
Auto		346	380	\$344	1%	329	5%	253	37%
OEM and other		88	79	\$80	10%	78	13%	66	33%
Total Revenue	\$28,000	\$30,040	\$29,992			\$26,044	15%	\$13,507	122%
COGS		7,311	7,348			5,484	33%	3,893	88%
Gross profit		22,729	22,644			20,560	11%	9,614	136%
Total Opex	\$2,800	2,792	2,909			2,501	12%	1,838	52%
Operating Income		19,937	19,735			18,059	10%	7,776	156%
Non-op (Expense) Income	\$300	380	302			302	26%	120	217%
Income tax (benefit) provision		3,365	3,408			3,123	8%	1,156	191%
Tax rate%	17.0%	17%	17%			17%	-45bps	15%	192bps
Net Income		16,952	16,629			15,238	11%	6,740	152%
Avg. Shares Out - Non-GAAP		24,848	24,890			24,890	0%	24,990	-1%
EPS - Non-GAAP		\$0.68	\$0.67			\$0.61	11%	\$0.27	153%

Sources: NVDA company reports, Thomson Reuters and Wedbush Securities, Inc. estimates

Income Statement and Other Selected Financials

NVDA.O	2024A				2025E				2026E								
	1Q24A	2Q24A	3Q24A	4Q24A	1Q25A	2Q25A	3Q25E	4Q25E	1Q26E	2Q26E	3Q26E	4Q26E					
\$Millions	Apr-23	Jul-23	Oct-23	Jan-24	Apr-24	Jul-24	Oct-24	Jan-25	Apr-25	Jul-25	Oct-25	Jan-26		Jan-23	Jan-24	Jan-25	Jan-26
EPS	\$0.11	\$0.27	\$0.40	\$0.52	\$0.61	\$0.68	\$0.73	\$0.76	\$0.84	\$0.93	\$0.99	\$1.02		\$0.33	\$1.30	\$2.78	\$3.78
Net Revenue	7,192	13,507	18,120	22,103	26,044	30,040	32,504	34,630	37,934	41,946	44,994	46,575		26,974	60,922	123,218	171,448
GM%	66.8%	71.2%	75.0%	76.7%	78.9%	75.7%	75.0%	74.5%	75.0%	75.0%	75.0%	75.0%		59.2%	73.8%	75.9%	75.0%
OM%	42.4%	57.6%	63.8%	66.7%	69.3%	66.4%	65.8%	65.1%	65.5%	65.3%	65.2%	65.0%		33.5%	61.0%	66.5%	65.2%
SEGMENT DATA	1Q24A	2Q24A	3Q24A	4Q24A	1Q25A	2Q25A	3Q25E	4Q25E	1Q26E	2Q26E	3Q26E	4Q26E		2023A	2024A	2025E	2026E
Gaming	2,240	2,486	2,856	2,865	2,647	2,880	3,027	3,008	3,044	3,312	3,391	3,309		9,067	10,447	11,563	13,056
Data Center	4,284	10,323	14,514	18,404	22,563	26,272	28,488	30,573	33,845	37,544	40,363	41,948		15,005	47,525	107,897	153,700
Professional visualization	295	379	416	463	427	454	520	532	470	499	572	586		1,544	1,553	1,933	2,127
Auto	296	253	261	281	329	346	392	422	494	473	587	632		903	1,091	1,488	2,186
OEM and other	77	66	73	90	78	88	77	95	82	117	80	99		455	306	337	379
Total Revenue	7,192	13,507	18,120	22,103	26,044	30,040	32,504	34,630	37,934	41,946	44,994	46,575		26,974	60,922	123,218	171,448
INCOME STATEMENT - Non-GAAP	1Q24A	2Q24A	3Q24A	4Q24A	1Q25A	2Q25A	3Q25E	4Q25E	1Q26E	2Q26E	3Q26E	4Q26E		2023A	2024A	2025E	2026E
Revenue	7,192	13,507	18,120	22,103	26,044	30,040	32,504	34,630	37,934	41,946	44,994	46,575		26,974	60,922	123,218	171,448
COGS	2,390	3,893	4,537	5,144	5,484	7,311	8,126	8,831	9,483	10,487	11,248	11,644		11,009	15,964	29,752	42,862
Gross profit	4,802	9,614	13,583	16,959	20,560	22,729	24,378	25,799	28,450	31,460	33,745	34,931		15,965	44,958	93,466	128,586
GM%	66.8%	71.2%	75.0%	76.7%	78.9%	75.7%	75.0%	74.5%	75.0%	75.0%	75.0%	75.0%		59.2%	73.8%	75.9%	75.0%
R&D	1,339	1,428	1,581	1,747	1,981	2,241	2,418	2,597	2,883	3,230	3,510	3,679		5,406	6,095	9,238	13,302
% revenue	18.6%	10.6%	8.7%	7.9%	7.6%	7.5%	7.4%	7.5%	7.6%	7.7%	7.8%	7.9%		20%	10%	7%	8%
SG&A	407	405	445	463	520	551	585	658	740	839	922	978		1,582	1,720	2,314	3,479
% revenue	5.7%	3.0%	2.5%	2.1%	2.0%	1.8%	1.8%	1.9%	2.0%	2.0%	2.1%	2.1%		6%	3%	2%	2%
Total Opex	1,750	1,838	2,026	2,210	2,501	2,792	3,003	3,255	3,623	4,069	4,432	4,657		6,925	7,824	11,552	16,781
% revenue	24.3%	13.6%	11.2%	10.0%	9.6%	9.3%	9.2%	9.4%	9.6%	9.7%	9.9%	10.0%		26%	13%	9%	10%
Operating Income	3,052	7,776	11,557	14,749	18,059	19,937	21,375	22,544	24,828	27,391	29,313	30,273		9,040	37,134	81,915	111,805
OM%	42.4%	57.6%	63.8%	66.7%	69.3%	66.4%	65.8%	65.1%	65.5%	65.3%	65.2%	65.0%		33.5%	61.0%	66.5%	65.2%
Non-op (Expense) Income	84	120	175	232	302	380	350	350	350	350	350	350		6	611	1,382	1,400
interest rate%	2.4%	2.5%	2.7%	2.7%	2.7%	2.7%	-	-	-	-	-	-		-	-	-	-
Pretax income	3,136	7,896	11,732	14,981	18,361	20,317	21,725	22,894	25,178	27,741	29,663	30,623		9,046	37,745	83,297	113,205
Income tax (benefit) provision	423	1,156	1,712	2,142	3,123	3,365	3,693	3,892	4,280	4,716	5,043	5,206		681	5,433	14,073	19,245
Tax rate%	13.5%	14.6%	14.6%	14.3%	17.0%	16.6%	17.0%	17.0%	17.0%	17.0%	17.0%	17.0%		7.5%	14.4%	16.9%	17.0%
Net Income	2,713	6,740	10,020	12,839	15,238	16,952	18,031	19,002	20,897	23,025	24,621	25,417		8,365	32,312	69,223	93,960
NM%	37.7%	49.9%	55.3%	58.1%	58.5%	56.4%	55.5%	54.9%	55.1%	54.9%	54.7%	54.6%		31.0%	53.0%	56.2%	54.8%
EPS - Non-GAAP	\$0.11	\$0.27	\$0.40	\$0.52	\$0.61	\$0.68	\$0.73	\$0.76	\$0.84	\$0.93	\$0.99	\$1.02		\$0.33	\$1.30	\$2.78	\$3.78
Avg. Shares Out - Non-GAAP	24,900	24,990	24,940	24,900	24,890	24,848	24,848	24,848	24,848	24,848	24,848	24,848		25,073	24,933	24,859	24,848
Net income - GAAP	2,043	6,188	9,243	12,285	14,881	16,599	17,515	18,478	20,540	22,550	24,104	24,893		4,368	29,759	67,473	92,088
GAAP EPS - Diluted	\$0.08	\$0.25	\$0.37	\$0.49	\$0.60	\$0.67	\$0.70	\$0.74	\$0.83	\$0.91	\$0.97	\$1.00		\$0.17	\$1.19	\$2.71	\$3.70
BALANCE SHEET	1Q24A	2Q24A	3Q24A	4Q24A	1Q25A	2Q25A	3Q25E	4Q25E	1Q26E	2Q26E	3Q26E	4Q26E		2023A	2024A	2025E	2026E
#days	91	91	91	91	91	91	91	91	91	91	91	91		364	364	364	364
Cash and cash equivalents	15,320	16,023	18,281	25,984	31,438	34,800	48,249	65,987	85,434	106,679	129,766	154,084		13,296	25,984	65,987	154,084
Accounts receivable, net	4,080	7,066	8,309	9,999	12,365	14,132	16,252	17,315	18,967	20,973	22,497	23,287		3,827	9,999	17,315	23,287
Inventories	4,611	4,319	4,779	5,282	5,864	6,675	9,751	10,389	11,380	12,584	13,498	13,972		5,159	5,282	10,389	13,972
Other current assets	872	1,389	1,289	3,080	4,062	4,026	4,026	4,026	4,026	4,026	4,026	4,026		791	3,080	4,026	4,026
Total current assets	24,883	28,797	32,658	44,345	53,729	59,633	78,278	97,717	119,807	144,262	169,787	195,369		23,073	44,345	97,717	195,369
Non-current assets:														-	-	-	-
PP&E, net	3,740	3,799	3,844	3,914	4,006	4,885	4,578	4,578	4,578	4,578	4,578	4,578		3,807	3,914	4,578	4,578
Investment in debt security	-	-	-	-	-	-	-	-	-	-	-	-		-	-	-	-
Goodwill	1,094	4,430	4,430	4,430	4,453	4,622	4,193	4,193	4,193	4,193	4,193	4,193		4,372	4,430	4,193	4,193
Other intangible assets, net	4,430	1,395	1,251	1,112	986	952	952	952	952	952	952	952		1,676	1,112	952	952
Other non-current assets	10,313	11,134	11,965	11,927	13,898	15,135	3,595	3,595	3,595	3,595	3,595	3,595		8,254	11,927	3,595	3,595
Total assets	44,460	49,555	54,148	65,728	77,072	85,227	91,596	111,035	133,125	157,580	183,105	208,687		41,182	65,728	111,035	208,687
Current liabilities:														-	-	-	-
Accounts payable	1,141	1,929	2,380	2,699	2,715	7,510	8,126	8,657	9,483	10,487	11,248	11,644		1,193	2,699	8,657	11,644
Accrued expenses	4,869	7,156	5,472	6,682	11,258	10,510	11,126	11,657	12,483	13,487	14,248	14,644		4,120	6,682	11,657	14,644
Current portion of LT debt	1,250	1,249	1,249	1,250	1,250	1,250	1,250	1,250	1,250	1,250	1,250	1,250		1,250	1,250	1,250	1,250
Total current liabilities	7,260	10,334	9,101	10,631	15,223	19,270	20,502	21,565	23,217	25,223	26,747	27,537		6,563	10,631	21,565	27,537
Non-current liabilities:														-	-	-	-
Long-term debt	9,704	8,456	8,457	8,459	8,460	8,460	8,460	8,460	8,460	8,460	8,460	8,460		9,703	8,459	8,460	8,460
Other liabilities	2,976	3,264	3,325	3,660	4,247	4,247	4,247	4,247	4,247	4,247	4,247	4,247		2,815	3,660	4,247	4,247
Total liabilities	19,940	22,054	20,883	22,750	27,930	31,977	33,209	34,272	35,924	37,930	39,454	40,244		19,081	22,750	34,272	40,244
Shareholders' Equity	24,520	27,501	33,265	42,978	49,142	53,250	58,387	76,763	97,201	119,649	143,651	168,443		22,101	42,978	76,763	168,443
Net (Debt) Cash	4,366	6,318	8,575	16,275	21,728	25,090	38,539	56,277	75,724	96,969	120,056	144,374		2,343	16,275	56,277	144,374
	\$ 0.25	\$ 0.34	\$ 0.65	\$ 0.87													
CASH FLOW	1Q24A	2Q24A	3Q24A	4Q24A	1Q25A	2Q25A	3Q25E	4Q25E	1Q26E	2Q26E	3Q26E	4Q26E		2023A	2024A	2025E	2026E
Net income - GAAP	2,043	6,188	9,24														

HARDWARE

TICKER	COMPANY NAME	WEDB RATING	PRICES	WEDB TP\$	UP/DNSIDE %	MKT CAP\$B	EV\$B	WEDB EPS EST				WEDB REV EST \$M				WEDB P/E				WEDB EV/SALES			
								2023A	2024E	2025E	2026E	2023A	2024E	2025E	2026E	2023A	2024E	2025E	2026E	2023A	2024E	2025E	2026E
2330	Taiwan Semiconductor Manufacturing Co Ltd	O	\$943.00	\$1,175.00	25%	25,000.0	23,972.8	\$32.32	\$41.32	\$54.65	\$65.89	\$2,161,740	\$2,766,160	\$3,366,850	\$3,944,210	27.9x	21.8x	16.5x	13.7x	10.8x	8.5x	6.9x	5.9x
AMD	Advanced Micro Devices Inc	O	\$146.36	\$200.00	37%	236.9	233.3	\$2.65	\$3.38	\$6.52	-	\$22,680	\$25,478	\$35,520	-	54.4x	42.6x	22.1x	-	10.3x	9.2x	6.6x	-
AXTI	AXT Inc	O	\$2.64	\$6.00	127%	0.1	0.2	(\$0.34)	(\$0.10)	\$0.46	-	\$76	\$106	\$149	-	-	-	-	-	-	-	-	-
DBD	Diebold Nixdorf Inc	O	\$43.12	\$50.00	16%	1.6	2.4	(\$0.44)	\$5.30	\$6.26	-	\$3,751	\$3,777	\$3,861	-	-98.0x	8.1x	6.9x	-	0.7x	0.6x	0.6x	-
GFS	GlobalFoundries Inc	O	\$45.35	\$65.00	43%	25.0	24.4	\$2.24	\$1.44	\$2.18	\$3.34	\$7,392	\$6,682	\$7,537	\$8,749	19.7x	30.6x	20.2x	13.2x	3.3x	3.6x	3.2x	2.8x
INTC	Intel Corp	N	\$19.61	\$25.00	27%	83.7	112.6	\$1.04	\$0.28	\$1.48	-	\$54,228	\$52,003	\$57,668	-	24.2x	89.9x	17.0x	-	2.0x	2.1x	1.9x	-
MU	Micron Technology Inc	O	\$94.86	\$170.00	79%	105.2	110.1	(\$4.45)	\$1.25	\$12.42	\$18.54	\$15,540	\$25,104	\$43,330	\$53,642	-22.3x	79.4x	8.0x	5.4x	7.1x	4.4x	2.5x	2.1x
NTAP	NetApp Inc	N	\$131.91	\$120.00	-9%	27.1	26.3	\$5.59	\$6.48	\$6.87	\$7.51	\$6,362	\$6,268	\$6,556	\$6,781	22.9x	19.7x	18.6x	17.0x	4.1x	4.2x	4.0x	3.9x
NVDA	NVIDIA Corp	O	\$125.61	\$138.00	10%	3090.0	3068.3	\$3.33	\$1.30	\$2.78	\$3.79	\$26,974	\$60,922	\$123,035	\$170,822	37.5x	95.9x	44.9x	32.9x	113.7x	50.3x	24.9x	18.0x
PSTG	Pure Storage Inc	O	\$59.76	\$70.00	17%	19.4	17.8	\$1.42	\$1.42	\$1.59	\$2.48	\$2,753	\$2,831	\$3,163	\$3,798	38.6x	38.6x	34.5x	22.1x	6.5x	6.3x	5.6x	4.7x
SIMO	Silicon Motion Technology Corp	O	\$62.10	\$90.00	45%	2.1	1.8	\$2.26	\$3.50	\$5.23	-	\$639	\$816	\$908	-	23.7x	15.3x	10.2x	-	2.9x	2.2x	2.0x	-
SMCI	Super Micro Computer Inc	N	\$443.49	\$620.00	40%	26.0	26.5	\$11.82	\$21.92	\$33.62	\$40.24	\$7,123	\$14,943	\$28,053	\$32,477	38.2x	20.6x	13.4x	11.2x	3.7x	1.8x	1.0x	0.8x
STX	Seagate Technology Holdings PLC	O	\$99.25	\$135.00	36%	20.8	25.2	\$0.19	\$1.28	\$8.93	\$13.21	\$7,359	\$6,552	\$9,980	\$12,018	630.5x	93.6x	13.4x	9.1x	3.4x	3.8x	2.5x	2.1x
2303	United Microelectronics Corp	O	\$54.60	\$60.00	10%	691.6	630.7	\$4.93	\$4.03	\$5.30	-	\$222,530	\$234,680	\$281,140	-	10.1x	12.3x	9.4x	-	2.8x	2.7x	2.2x	-
WDC	Western Digital Corp	O	\$62.43	\$95.00	52%	21.4	27.2	(\$1.86)	(\$0.24)	\$10.01	\$14.15	\$12,318	\$13,003	\$18,257	\$21,907	-42.7x	-330.9x	7.9x	5.6x	2.1x	2.0x	1.4x	1.2x

TICKER	COMPANY NAME	MEAN REC	PRICES	MEAN TP\$	UP/DNSIDE %	MKT CAP\$B	EV\$B	MEAN EPS EST				MEAN REV EST \$M				MEAN P/E				MEAN EV/SALES			
								2023A	2024E	2025E	2026E	2023A	2024E	2025E	2026E	2023A	2024E	2025E	2026E	2023A	2024E	2025E	2026E
2330	Taiwan Semiconductor Manufacturing Co Ltd	O	\$943.00	\$1,241.03	32%	25,000.0	23,972.8	\$32.34	\$41.81	\$53.59	\$64.06	\$2,161,736	\$2,809,170	\$3,470,539	\$4,094,307	27.9x	21.6x	16.8x	14.1x	10.8x	8.3x	6.7x	5.7x
AMD	Advanced Micro Devices Inc	O	\$146.36	\$186.63	28%	236.9	233.3	\$2.65	\$3.40	\$5.43	\$7.36	\$22,680	\$25,607	\$32,875	\$39,820	54.4x	42.4x	26.6x	19.6x	10.3x	9.1x	7.1x	5.9x
AXTI	AXT Inc	O	\$2.64	\$5.50	108%	0.1	0.2	(\$0.42)	(\$0.18)	\$0.09	-	\$76	\$104	\$127	-	-	-	-	-	-	-	-	-
DBD	Diebold Nixdorf Inc	O	\$43.12	\$57.50	33%	1.6	2.4	\$2.84	\$5.30	\$5.83	-	\$3,761	\$3,756	\$3,812	-	22.7x	12.2x	11.0x	-	0.7x	0.7x	0.6x	-
GFS	GlobalFoundries Inc	N	\$45.35	\$55.46	22%	25.0	24.4	\$2.24	\$1.36	\$1.90	\$2.75	\$7,392	\$6,725	\$7,625	\$8,588	19.7x	32.3x	23.2x	16.0x	3.3x	3.6x	3.2x	2.8x
INTC	Intel Corp	N	\$19.61	\$26.47	35%	83.7	112.6	\$1.05	\$0.27	\$1.14	\$1.83	\$54,228	\$52,317	\$56,414	\$60,808	24.0x	94.0x	22.1x	13.7x	2.0x	2.1x	1.9x	1.8x
MU	Micron Technology Inc	O	\$94.86	\$171.24	81%	105.2	110.1	(\$4.45)	\$1.22	\$9.39	\$13.10	\$15,540	\$25,062	\$38,464	\$44,804	-22.3x	81.5x	10.6x	7.6x	7.1x	4.4x	2.9x	2.5x
NTAP	NetApp Inc	N	\$131.91	\$128.70	-2%	27.1	26.3	\$5.59	\$6.46	\$6.92	\$7.48	\$6,362	\$6,268	\$6,558	\$6,879	22.9x	19.8x	18.5x	17.1x	4.1x	4.2x	4.0x	3.8x
NVDA	NVIDIA Corp	O	\$125.61	\$139.97	11%	3090.0	3068.3	\$0.33	\$1.30	\$2.77	\$3.84	\$26,974	\$60,922	\$122,601	\$170,219	37.3x	96.2x	45.0x	32.5x	113.7x	50.3x	25.0x	18.0x
PSTG	Pure Storage Inc	N	\$59.76	\$69.48	16%	19.4	17.8	\$1.41	\$1.42	\$1.60	\$1.88	\$2,753	\$2,831	\$3,133	\$3,543	38.8x	38.6x	34.2x	29.2x	6.5x	6.3x	5.7x	5.0x
SIMO	Silicon Motion Technology Corp	N	\$62.10	\$87.00	40%	2.1	1.8	\$2.27	\$3.48	\$4.74	\$5.94	\$639	\$817	\$892	\$1,019	23.6x	15.4x	11.3x	9.0x	2.9x	2.2x	2.0x	1.8x
SMCI	Super Micro Computer Inc	N	\$443.49	\$858.46	94%	26.0	26.5	\$11.81	\$22.09	\$34.62	\$44.93	\$7,123	\$14,943	\$28,371	\$33,615	38.3x	20.5x	13.1x	10.1x	3.7x	1.8x	0.9x	0.8x
STX	Seagate Technology Holdings PLC	N	\$99.25	\$115.85	17%	20.8	25.2	\$0.19	\$1.29	\$6.88	\$9.05	\$7,384	\$6,551	\$9,151	\$10,361	630.5x	92.9x	17.4x	13.2x	3.4x	3.8x	2.7x	2.4x
2303	United Microelectronics Corp	N	\$54.60	\$57.04	4%	691.6	630.7	\$4.83	\$4.16	\$4.78	\$5.01	\$222,533	\$233,860	\$265,329	\$294,848	10.3x	11.9x	10.4x	9.9x	2.8x	2.7x	2.3x	2.1x
WDC	Western Digital Corp	N	\$62.43	\$90.13	44%	21.4	27.2	(\$3.59)	(\$0.20)	\$8.04	\$8.83	\$12,318	\$13,003	\$17,395	\$19,092	-22.1x	-397.1x	9.9x	9.0x	2.1x	2.0x	1.5x	1.4x

Source: Thomson Reuters & Wedbush Securities, Inc. estimates; Priced: 08/28/2024

* All numbers for 2330.TW and 2303.TW in NTS

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VALUATION

2330	We arrive at our target price by applying a PE multiple of ~17X to our FY2026 EPS estimate (plus net cash).
AMD	We arrive at our target price by applying a PE multiple of ~30X to our FY2025 EPS estimate (plus net cash).
AXTI	We arrive at our target price by applying a PE multiple of ~13X to our FY2025 EPS estimate (plus net debt).
DBD	We arrive at our target price by applying a PE multiple of ~8X to our FY2025 EPS estimate.
GFS	We arrive at our target price by applying a PE multiple of ~19X to our FY2026 EPS estimate (plus net cash).
INTC	We arrive at our target price by applying a multiple of ~20X to our FY2025 (net debt) outlook.
MU	We arrive at our target price by applying a PE multiple of ~9X to our FY2026 (plus net debt) outlook.
NTAP	We arrive at our target price by applying a PE multiple of ~15X to our FY2026 estimates and adding net cash.
NVDA	We arrive at our target price by applying a PE multiple of ~36X to our FY2026 EPS estimate (plus net cash).
PSTG	We arrive at our target price by applying a PE multiple of ~29X to our FY2026 estimates (plus net cash).
SIMO	We arrive at our target price by applying a PE multiple of ~12X to our FY2025 est + net cash + assumed MXL damages at least equal to the prior termination fee.
SMCI	We arrive at our target price by applying a PE multiple of ~15X to our FY2026 EPS (net debt).
STX	We arrive at our target price by applying a PE multiple of ~12X to our FY2026 estimate (plus net debt).
2303	We arrive at our target price based on a multiple of ~9X PE using our FY2025 EPS estimate (net cash).
WDC	We arrive at our target price by applying a PE multiple of ~8X to our FY2026 estimate (plus net debt).

RISKS

2330	Economic deterioration and demand destruction related to the pandemic, changes to US Policy, sovereignty risk to Taiwan, risk to the supply chain, and technology risk.
AMD	AMD misexecutes, customers choose other vendors, competition increases (e.g., pricing, design), TSMC is constrained (production), the economy deteriorates.
AXTI	Demand for AXTI products fall; Tongmei fails to IPO; China/US trade risk; market share loss.
DBD	Demand for DBD's products falls, competition increases pricing pressure, fails to satisfy debt service, or fails to improve on working capital linearity.
GFS	Economic deterioration and demand destruction related to the pandemic, changes to government policy, risk to the supply chain, and technology risk.
INTC	Upside risk: INTC executes on stated initiatives, competition misexecutes, tailwinds continue to boost revenues and GM. Downside risk: misexecution, recovery elongates
MU	MU misexecutes, future acquisitions dilute value, Chinese vendors successfully enter NAND or DRAM markets, US/China trade war.
NTAP	Upside risks: Successful product innovation (software), competitor missteps and accelerated TAM growth. Downside risks: Cloud cannibalization, normalized enterprise spend, increasing competition and misexecution.
NVDA	DC weakens; AI driven demand softens; gaming demand softens, misexecution, customers reduce spending, competition gains share, US/China tensions increase.
PSTG	Downside risks: Cloud cannibalization, normalization/decline in enterprise spend, increasing competition and misexecution.
SIMO	MXL could successfully terminate its proposed acquisition of SIMO without compensating SIMO, misexecution.
SMCI	SMCI out executes/innovates (misexecutes/fails to innovate) vs. peers, demand increases (decreases).
STX	STX fails to execute on stated initiatives and particularly HAMR, pricing decreases, demand decreases, and US/China tensions increase.
2303	Economic deterioration and demand destruction related to the pandemic, changes to US Policy, sovereignty risk to Taiwan, risk to the supply chain, and technology risk.
WDC	WDC misexecutes, customers choose other vendors, competition increases (e.g., pricing, design), the economy deteriorates. The ann'd split of WDC's HDD and NAND fails to materialize.

Valuation

We arrived at our price target by applying a PE multiple of ~36X to our FY2026 estimate (plus net cash).

Company Description

NVDA is a fabless semiconductor company that designs and sells GPUS for gaming, data center, and other markets.

Risks to the Attainment of Our Price Target and Rating:

Potential acquisitions; Data Center customer investment softens; AI demand deteriorates; government regulations weigh on the growth of AI; gaming demand remains weak; competitors (e.g., AMD, INTC etc.) are successful in taking share in AI silicon; customers shift to internally developed products; tensions between the West and China grow culminating in NVDA losing share in that market and/or disruption to NVDA's supply chain; NVDA misexecutes on future product development or roll-outs.

Analyst Certification

We, Matt Bryson and Antoine Legault, certify that the views expressed in this report accurately reflect our personal opinions and that we have not and will not, directly or indirectly, receive compensation or other payments in connection with our specific recommendations or views contained in this report.

Mentioned Companies

Company	Rating	Price	Target
United Microelectronics	OUTPERFORM	TWD50.90	TWD60.00
Taiwan Semiconductor	OUTPERFORM	TWD934.00	TWD1,175.00
Advanced Micro Devices Inc	OUTPERFORM	\$155.28	\$200.00
Amazon.com	OUTPERFORM	\$171.40	\$225.00
AXT Inc.	OUTPERFORM	\$3.29	\$6.00
Intel Corp	NEUTRAL	\$29.05	\$25.00
Micron Technology Inc	OUTPERFORM	\$142.36	\$170.00
NetApp Inc.	NEUTRAL	\$125.47	\$120.00
Nvidia	OUTPERFORM	\$125.61	\$138.00
Pure Storage Inc.	OUTPERFORM	\$67.68	\$80.00
Silicon Motion Technology Corp	OUTPERFORM	\$62.04	\$90.00
Super Micro Computer, Inc.	NEUTRAL	\$616.94	\$620.00
Seagate Technology Holdings PLC	OUTPERFORM	\$105.30	\$135.00
Western Digital	OUTPERFORM	\$61.51	\$95.00
Meta Platforms Inc	OUTPERFORM	\$508.87	\$600.00
Alphabet	OUTPERFORM	\$159.25	\$205.00
Microsoft	OUTPERFORM	\$422.92	\$550.00

Investment Rating System:

OUTPERFORM: Expect the total return of the stock to outperform relative to the median total return of the analyst's (or the analyst's team) coverage universe over the next 6-12 months.

NEUTRAL: Expect the total return of the stock to perform in-line with the median total return of the analyst's (or the analyst's team) coverage universe over the next 6-12 months.

UNDERPERFORM: Expect the total return of the stock to underperform relative to the median total return of the analyst's (or the analyst's team) coverage universe of the next 6-12 months.

The Investment Ratings are based on the expected performance of a stock (based on anticipated total return to price target) relative to the other stocks in the analyst's coverage universe (or the analyst's team coverage).*

Rating distribution (as of August 29, 2024)	Investment Banking Relationships (as of August 29, 2024)
OUTPERFORM: 59.80% NEUTRAL: 37.72% UNDERPERFORM: 2.48%	OUTPERFORM: 8.71% NEUTRAL: 3.95% UNDERPERFORM: 20.00%

The Distribution of Ratings is required by FINRA rules; however, WS' stock ratings of Outperform, Neutral, and Underperform most closely conform to Buy, Hold, and Sell, respectively. Please note, however, the definitions are not the same as WS' stock ratings are on a relative basis.

The analysts responsible for preparing research reports do not receive compensation based on specific investment banking activity. The analysts receive compensation that is based upon various factors including WS' total revenues, a portion of which are generated by WS' investment banking activities.

Company Specific Disclosures

This information is subject to change at any time.

1. WS makes a market in the securities of Nvidia, United Microelectronics, Taiwan Semiconductor, Advanced Micro Devices Inc, Amazon.com, AXT Inc., Alphabet, Intel Corp, Meta Platforms Inc, Microsoft, Micron Technology Inc, NetApp Inc., Pure Storage Inc., Silicon Motion Technology Corp, Super Micro Computer, Inc., Seagate Technology Holdings PLC and Western Digital.

10. The research analyst, a member of the research analyst's household, any associate of the research analyst, or any individual directly involved in the preparation of this report has a long position in the common stocks of Amazon.com, Alphabet and Microsoft.

Price Charts

Wedbush disclosure price charts are updated within the first fifteen days of each new calendar quarter per FINRA regulations. Price charts for companies initiated upon in the current quarter, and rating and target price changes occurring in the current quarter, will not be displayed until the following quarter. Additional information on recommended securities is available on request.

Disclosure information regarding historical ratings and price targets is available: [Research Disclosures](#)

*WS changed its rating system from (Strong Buy/ Buy/ Hold/ Sell) to (Outperform/ Neutral/ Underperform) on July 14, 2009.

Applicable disclosure information is also available upon request by contacting the Research Department at (212) 833-1375, by email to leslie.lippai@wedbush.com. You may also submit a written request to the following: Wedbush Securities, Attn: Research Department, 142 W 57th Street, New York, NY 10019.

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Retail Investors

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